

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1023	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	N. Lahari	Reg. No.:	192473018
Date:	06-10-26	Duration:	60 minutes

Code of Conduct

I N. Lahari (192473018)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Lahari_____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

CASE STUDY ANALYSIS

Problem Statement No.: 26

Title: AI-Assisted Customer Support with Human Escalation Control

1. Problem Statement

Design and develop an AI-Assisted Customer Support Platform that automatically handles routine customer requests while identifying cases that require human intervention.

The platform should maintain conversation context, integrate secure APIs, classify requests, detect uncertainty, escalate sensitive cases, and preserve audit records. It should also support scalable architecture, testing, CI/CD, monitoring, and responsible AI practices.

2. Industry Context

Organizations are increasingly using AI assistants to handle customer requests and provide quick responses. AI can effectively manage routine questions such as frequently asked questions, order status, and basic product information.

However, incorrect responses, ambiguous requests, privacy-sensitive questions, and high-impact decisions may require human intervention. Therefore, the proposed system combines AI automation with human supervision. Routine requests can be handled automatically, while uncertain, sensitive, or high-risk cases are transferred to human support staff.

3. Problem Understanding

The problem is to design an AI-Assisted Customer Support Platform that can automatically handle routine customer requests while ensuring that uncertain, sensitive, and high-risk cases are transferred to human support staff.

The main challenge is to balance **automation and human supervision**. Although AI can respond quickly to common customer queries, it may sometimes provide incorrect or uncertain responses. Some customer requests may also involve private information, complaints, account security, or decisions that require human judgment.

Therefore, the proposed system must first understand and classify every customer request. Based on the request type, risk level, and confidence of the AI system, it should either generate an automatic response or escalate the conversation to a human support agent.

The system should also maintain the conversation history so that the customer does not need to repeat the

same information after escalation. Secure APIs should be used to access required services and customer information.

System Constraints

The system should:

1. Protect customer privacy and sensitive information.
2. Avoid providing unreliable answers when AI confidence is low.
3. Escalate high-risk and sensitive cases to humans.
4. Maintain conversation context.
5. Provide secure API access.
6. Support a large number of customer requests.
7. Maintain proper audit records.
8. Follow responsible AI principles such as fairness, transparency, and accountability.

4. Objectives

The major objectives of the proposed system are:

1. To automatically handle routine customer requests.
2. To classify customer requests according to their type and risk level.
3. To detect uncertainty in AI-generated responses.
4. To identify sensitive and high-impact customer cases.
5. To automatically escalate high-risk cases to human support staff.
6. To maintain complete conversation context during customer interactions.
7. To integrate secure APIs with external customer and service systems.
8. To maintain audit records of important interactions and decisions.
9. To monitor response quality and system performance.
10. To provide a scalable and reliable customer-support solution.
11. To ensure privacy, fairness, transparency, and ethical use of AI.

5. Proposed System

The proposed system is an **AI-Assisted Customer Support Platform with Human Escalation Control**. The system combines artificial intelligence with human support to provide fast, reliable, and secure customer service.

The system receives a customer query through the customer-support interface. It first checks the customer's information and retrieves the relevant conversation history. The query is then processed by the AI system and classified according to its type and risk level.

Routine and low-risk requests are handled automatically by the AI assistant. However, if the system detects

low confidence, sensitive information, a complaint, a security-related issue, or a high-impact decision, the request is transferred to a human support agent.

The human agent receives the complete conversation context and can review the AI's analysis before providing the final response. The interaction is then stored in the audit system for monitoring and future improvement.

5.1 Main Components of the System

1. Customer Interface

Provides an interface through which customers can submit questions, complaints, requests, and other support-related messages.

2. Conversation Management

Maintains the customer's previous messages and conversation history. This allows the system to understand the context of the current request.

3. AI Query Processing

Analyses the customer's message using AI and identifies the intent, category, and required action.

4. Risk and Uncertainty Detection

Evaluates the AI response and determines whether the request is safe to automate. Low-confidence or high-risk cases are identified for escalation.

5. Human Escalation

Transfers sensitive, uncertain, or high-risk cases to a human support agent.

6. Secure API Layer

Provides secure communication between the customer-support platform and external services such as customer databases, order systems, authentication services, and notification services.

7. Audit and Monitoring

Records important system activities, AI decisions, escalations, and customer-support interactions for monitoring and accountability.

6. Algorithm Design

The following algorithm describes how the proposed customer-support system operates.

Algorithm: AI-Assisted Customer Support with Human Escalation

Input: Customer query, customer details, and conversation history.

Output: AI-generated response or human-supported response.

Steps:

1. Start the customer-support system.
2. Receive the customer query.
3. Authenticate and identify the customer when required.

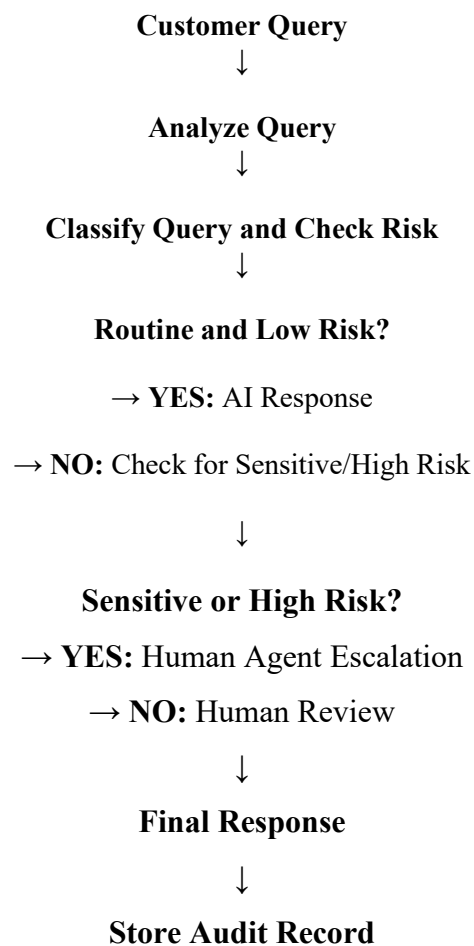
4. Retrieve the previous conversation history.
5. Analyse the customer query using the AI processing module.
6. Identify the intent and category of the query.
7. Calculate the AI confidence level.
8. Determine the risk and sensitivity level of the request.
9. Check whether the query is routine and safe to automate.
10. If the query is routine and the confidence level is high, generate an AI response.

7. Decision-Making Logic

The system uses conditional statements to decide whether a customer query should be handled by AI or transferred to a human agent.

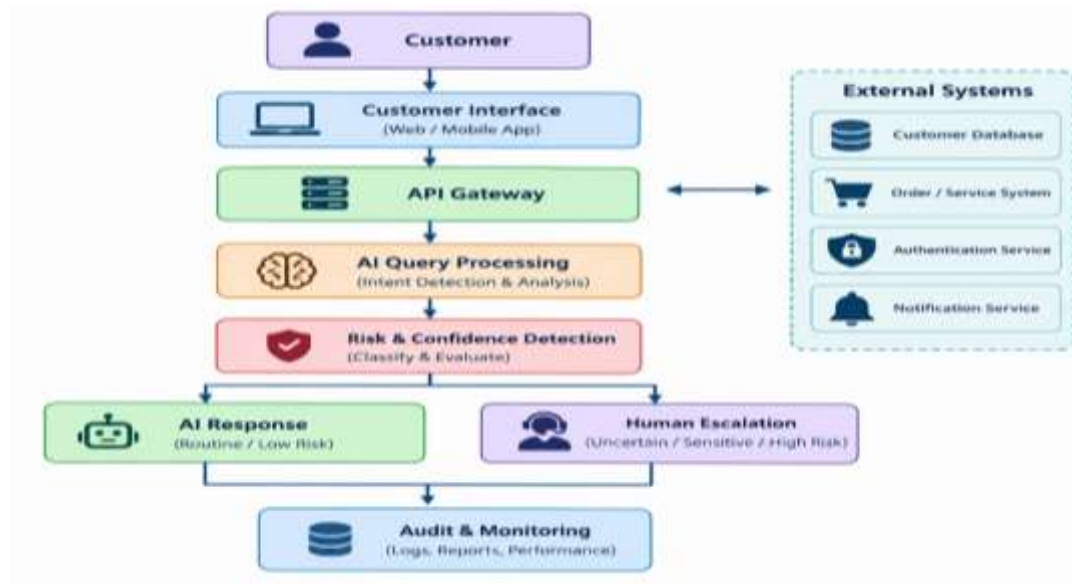
- **Routine + Low Risk + High Confidence** → AI handles the query.
- **Uncertain Query** → Human review.
- **Sensitive/High-Risk Query** → Immediate human escalation.
- The final response is stored in the audit record

7.1 Decision Flow



9. System Architecture

The proposed system follows a layered architecture in which customer requests pass through different modules before a final response is provided.



10. Security and Privacy

Security and privacy are important because the system handles customer information and conversation records.

- Secure authentication is used to protect customer accounts.
- APIs are protected to prevent unauthorized access.
- Sensitive customer information is handled carefully.
- Only authorized support agents can access required customer details.
- Conversation history and audit records are securely maintained.

11. Testing and Validation

The system is tested with different types of customer queries to check whether the correct action is taken.

1. **Routine Query:** AI provides an automatic response.
2. **Uncertain Query:** Query is sent for human review.
3. **Sensitive Query:** Query is escalated to a human agent.
4. **High-Risk Query:** Immediate human intervention is provided.
5. **API Failure:** System should handle the error safely without losing customer data.

Testing ensures that the system provides reliable responses and performs proper escalation when required.

12. Scalability and Monitoring

The system can support many customer requests using scalable cloud-based services. Monitoring helps track response time, AI accuracy, errors, and the number of escalated cases. These records can be used to improve

system performance.

13. Expected Outcome

The system reduces the workload of customer-support agents by automatically handling routine requests. At the same time, human agents can focus on sensitive, uncertain, and high-risk cases, resulting in faster and safer customer support.

14. Responsible AI Practices

The system should use AI in a safe, fair, and transparent manner.

- **Fairness:** Customer requests should be handled without discrimination.
- **Transparency:** Important AI decisions should be understandable.
- **Privacy:** Customer data should be protected.
- **Accountability:** Human agents should be responsible for important decisions.
- **Reliability:** Low-confidence AI responses should be escalated instead of giving incorrect information.

15. Advantages

1. Provides quick responses to routine customer queries.
2. Reduces workload on human support agents.
3. Automatically identifies risky and sensitive requests.
4. Maintains complete conversation context.
5. Improves customer service efficiency.
6. Provides audit records for monitoring and analysis.

16. Future Enhancements

The system can be enhanced by improving AI accuracy, supporting multiple languages, adding voice-based customer support, and using advanced analytics to identify common customer problems.

17. Conclusion

The AI-Assisted Customer Support Platform combines AI automation with human supervision. It provides efficient support for routine queries while ensuring that uncertain, sensitive, and high-risk cases are handled by human agents. This makes the system reliable, secure, and suitable for modern customer-support environments.

18. Result / Expected System Output

The proposed system processes customer queries and takes appropriate action based on the query type, risk level, and AI confidence.

Query Type	System Action
Routine Query	AI Response

Query Type	System Action
Low Confidence	Human Review
Sensitive Query	Human Escalation
High-Risk Query	Immediate Escalation

19. Overall Outcome

The proposed platform provides a balance between AI automation and human supervision. Routine requests are handled quickly by AI, while sensitive and uncertain cases are safely transferred to human agents. This improves customer satisfaction, reduces support workload, and ensures responsible use of AI.

20. Implementation Technologies

The proposed system can be implemented using the following technologies:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python / Java
- **AI Processing:** Natural Language Processing and Machine Learning
- **Database:** MySQL / MongoDB
- **API:** REST APIs for secure communication

21. Key Performance Measures

The system performance can be evaluated using:

1. **Response Time** – Time taken to respond to customer queries.
2. **AI Accuracy** – Correctness of automatically generated responses.
3. **Escalation Accuracy** – Correct identification of cases requiring human support.
4. **System Availability** – Ability of the platform to remain available for customers.
5. **Customer Satisfaction** – Quality and usefulness of the support provided

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system,	Good analysis with most strengths and	Limited analysis with partial	Proposed solution is unclear or inappropriate;
	strengths, weaknesses, and identifies all gaps.	weaknesses identified.	identification of issues.	requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/ framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

